

06b - Multi-Contour Handoff and Tool Lease Fixture Pack v0.1

Academic PDF rendering for the CCALC / C-calculus governed binding stack v0.1.

Document ID	06b_MULTI_CONTOUR_HANDOFF_AND_TOOL_LEASE_FIXTURE_PACK_v0_1
Version/status	executable fixture pack / mutation hardening layer for 06a
Stack layer	multi-contour handoff and tool lease
Date	2026-07-05T09:08:00Z
Author	Kotov Ivan, Bruxelles, Belgique - 2026
ORCID	0009-0009-6002-9845
Reserved DOI	10.5281/zenodo.21205427
Notice	This DOI is reserved and becomes active after Zenodo publication.

Abstract / purpose

06b hardens the most operationally dangerous part of the runtime authority boundary: 06 states the normative boundary. 06a provides schema/checker seed records. 06b adds adversarial fixture coverage and mutation tests for the practical failure modes where one contour tries to borrow another contour's authority, memory root, witness root, owner anchor, or tool credential. The core rule remains: Operational expansion: ---

Non-claims boundary

- _ not legal advice
- _ not privacy-law certification
- _ not safety certification
- _ not deployment authorization
- _ not standards compliance certification
- _ not regulated approval
- _ not C-A1 ratification
- _ not personhood proof
- _ not consciousness proof
- _ not live substrate truth
- _ not proof of completeness

06b - Multi-Contour Handoff and Tool Lease Fixture Pack v0.1

Artifact: 06b_MULTI_CONTOUR_HANDOFF_AND_TOOL_LEASE_FIXTURE_PACK_v0_1 Package:
CCALC_MULTI_CONTOUR_HANDOFF_TOOL_LEASE_06b_v0_1 Status: executable fixture pack / mutation hardening
layer for 06a Created UTC: 2026-07-05T09:08:00Z Review mode: direct construction; no external b-layer reviewer
record included.

0. Purpose

06b hardens the most operationally dangerous part of the runtime authority boundary:

cross-contour handoff + tool surface lease

06 states the normative boundary. 06a provides schema/checker seed records. 06b adds adversarial fixture
coverage and mutation tests for the practical failure modes where one contour tries to borrow another contour's
authority, memory root, witness root, owner anchor, or tool credential.

The core rule remains:

shared infrastructure creates risk, not authority.

Operational expansion:

handoff != command
tool lease != owner approval
source evidence != target admission
source credential != target credential
foreign witness != target witness completeness
shared host != shared identity
shared tool broker != shared authority

1. Source bindings

Binding	File	SHA-256	Role
DOC04_CONTINUITY_STACK_UMBRELLA	CCALC_DOC04_CONTINUITY_STACK_UMBRELLA_v0_1.zip	6caeab8d489aaa6b26902db3bd3e4c5169e5c8d5617b93aed6586ea900322d	continuity classification dependency
DOC05_SELF_EVO_STACK_UMBRELLA	CCALC_DOC05_SELF_EVO_STACK_UMBRELLA_v0_1.zip	9a0717809029f49df9001dc8ffa1803d9e9bfa1a824d317eebb146cbc7df43b4	self-evolution/promotion/watch dependency
DOC06_RUNTIME_AUTHORITY_BOUNDARY	CCALC_RUNTIME_AUTHORITY_BOUNDARY_06_v0_1.zip	863d187d9631214bd891ffaf262d869305a11e52752944f8b110418b9d81dff	normative runtime authority boundary
DOC06A_RUNTIME_AUTHORITY_MANIFEST	CCALC_RUNTIME_AUTHORITY_MANIFEST_06a_v0_1.zip	30ff69a365dfcedbb80a0fd661bd45bd9e9d7edd02546782cac8081a3fb96ad6	schema/checker parent dependency

A missing or mismatched binding makes the fixture pack unsuitable for release use.

2. Record surfaces exercised

06b focuses on three executable drill families:

CrossContourHandoffDrill
ToolLeaseDrill
CombinedHandoffToolLeaseDrill

These are not replacement schemas for 06a; they are test surfaces used to pressure the 06a semantics.

3. Guard set

The fixture pack enforces and mutates the following boundary guards:

B06b-01 handoff is not command
B06b-02 receiver admission required
B06b-03 source authority cannot approve target effect
B06b-04 identity/authority pressure denied
B06b-05 core memory import requires target anchor and gate
B06b-06 UNKNOWN continuity cannot authorize persistent effect
B06b-07 negative-cache HIT cannot ALLOW
B06b-08 unresolved red pattern cannot ALLOW
B06b-09 tool wildcard operation denied
B06b-10 shared credential cannot launder authority
B06b-11 revoke path must exist and be tested
B06b-12 L4/external tool requires owner anchor
B06b-13 borrowed authority/credential denied

4. Intended reading against 06/06a

06b is intentionally narrow. It does not define new runtime rights. It demonstrates that the runtime boundary rejects the common dangerous shortcuts:

```
source says execute -&gt; target treats as request only
source says we are same -&gt; target denies identity pressure
source credential forwarded -&gt; target denies borrowed authority
green source test -&gt; target still requires admission
foreign memory proposed -&gt; target memory gate decides
shared credential -&gt; per-contour attribution or denial
wildcard lease -&gt; denial
no revoke path -&gt; hold/denial
UNKNOWN continuity + persistent effect -&gt; hold/denial
```

5. Fixture scope

The package contains fixtures across:

```
valid handoff variants
invalid handoff variants
valid tool lease variants
invalid tool lease variants
valid combined handoff/tool lease variants
invalid combined handoff/tool lease variants
mutation-target one-fault adversarial records
```

The runner treats fixture sidecars as non-input and loads only *.json cases.

6. Mutation scope

The mutation harness disables one guard at a time and re-runs a targeted one-fault case. A mutation is considered caught when the disabled guard would make an otherwise invalid record pass.

Mutation classes:

```
MUT_ALLOW_COMMAND_HANDOFF
MUT_IGNORE_RECEIVER_ADMISSION
MUT_ALLOW_SOURCE_AUTHORITY
MUT_ALLOW_IDENTITY_PRESSURE
MUT_ALLOW_TARGET_MEMORY_CORE_NO_ANCHOR
MUT_IGNORE_UNKNOWN_CONTINUITY
MUT_IGNORE_NEGATIVE_CACHE
MUT_IGNORE_RED_PATTERN
MUT_ALLOW_TOOL_WILDCARD
MUT_ALLOW_SHARED_CREDENTIAL
MUT_IGNORE_TOOL_REVOKE
MUT_IGNORE_L4_ANCHOR
MUT_ALLOW_BORROWED_AUTHORITY
```

7. Non-claims

06b is not a safety certification, deployment authorization, C-A1 ratification, proof of live substrate truth, or proof of completeness. It is a fixture/mutation hardening pack for the handoff and lease boundary.

8. Acceptance condition

```
python3 scripts/run_06b_fixtures.py
python3 scripts/run_06b_mutations.py
sha256sum -c SHA256SUMS.txt
```

All three must pass for the package to be treated as sealed.